

Lab Title: Case Study 3 Rhino Hunting / Illegal Possession Investigation

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Student ID: 2026/DFIT/17036

Course Name: Computer Forensics Case Study I (CIP-B104)

Date of Submission: 9/12/2026

Instruction Name: Aminu Idris

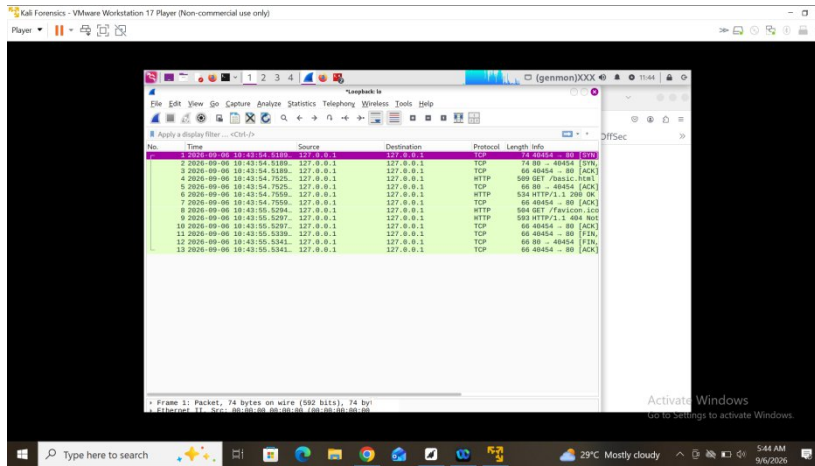
Platform Used: ICDFA Kali Linux and Wireshark

Executive Summary

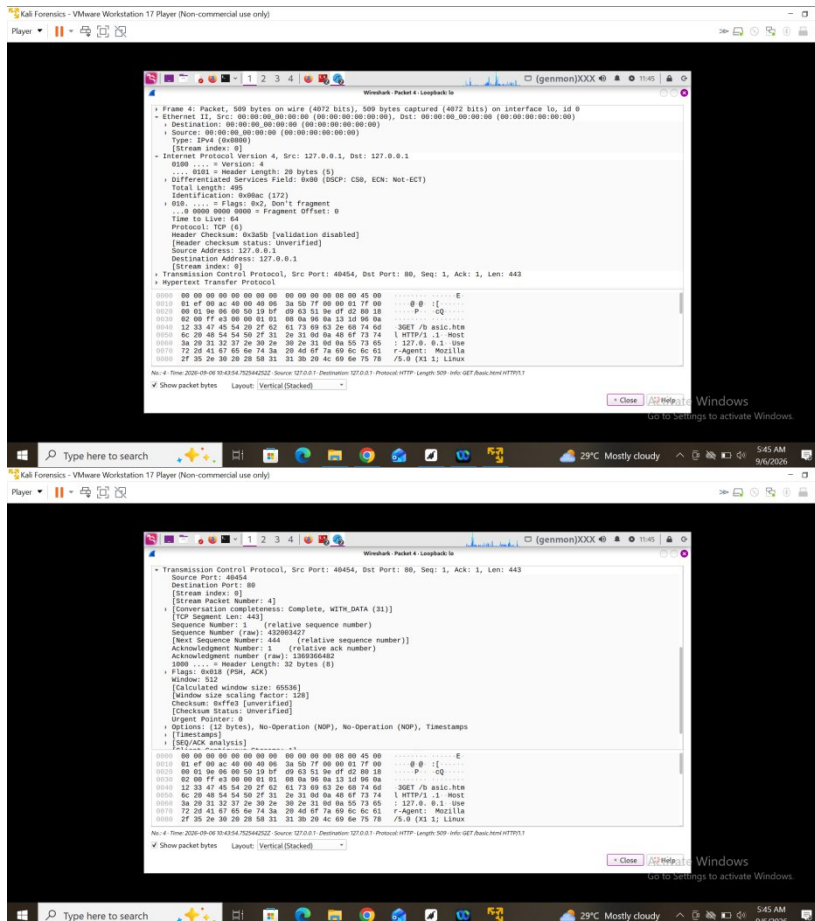
This investigation examined USB, FTP, HTTP, steganographic, archive, and executable evidence to identify rhinoceros images. Original evidence was preserved, while recovered files were validated, hashed, and compared to eliminate duplicates. The investigation established the provenance and movement of relevant images across multiple evidence sources, with findings documented in a structured timeline and provenance record. The analysis also considered evidence limitations and distinguished evidence of possession or transfer from conclusions about who performed the actions.

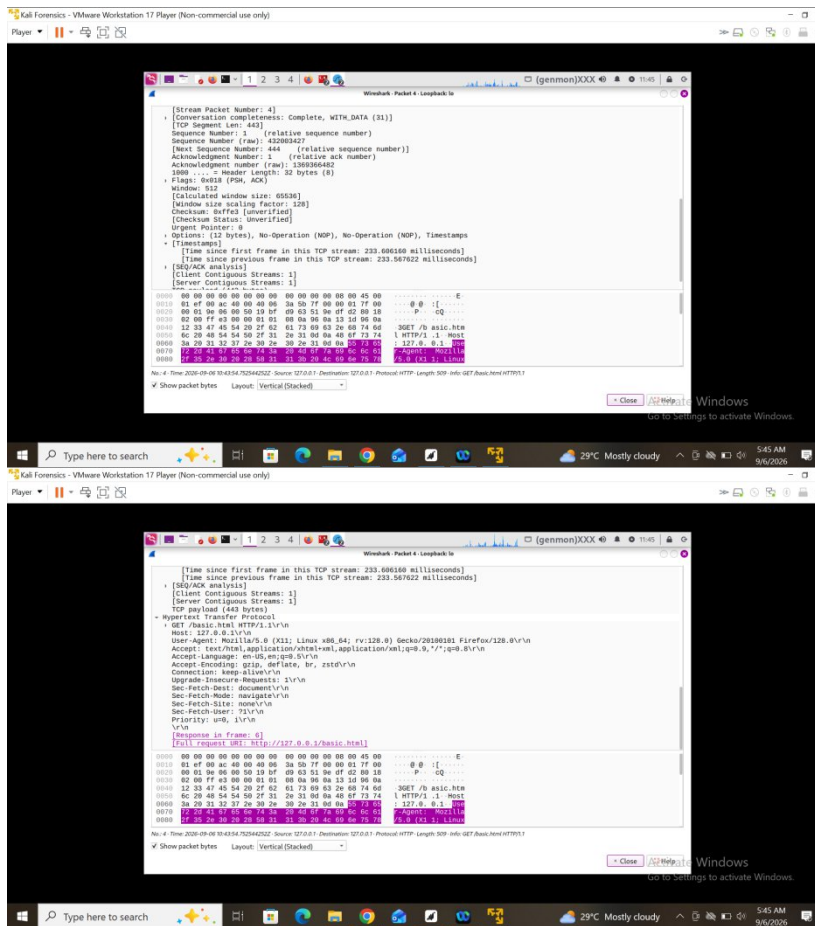
HTTP Wireshark Forensic Text

1. The captured http traffic using Wireshark



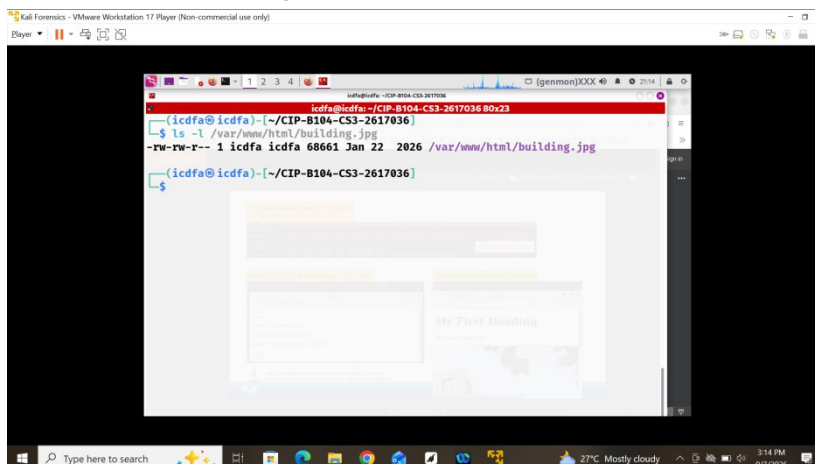
2. Analyzing the traffic by looking t the request, response and tcp protocol

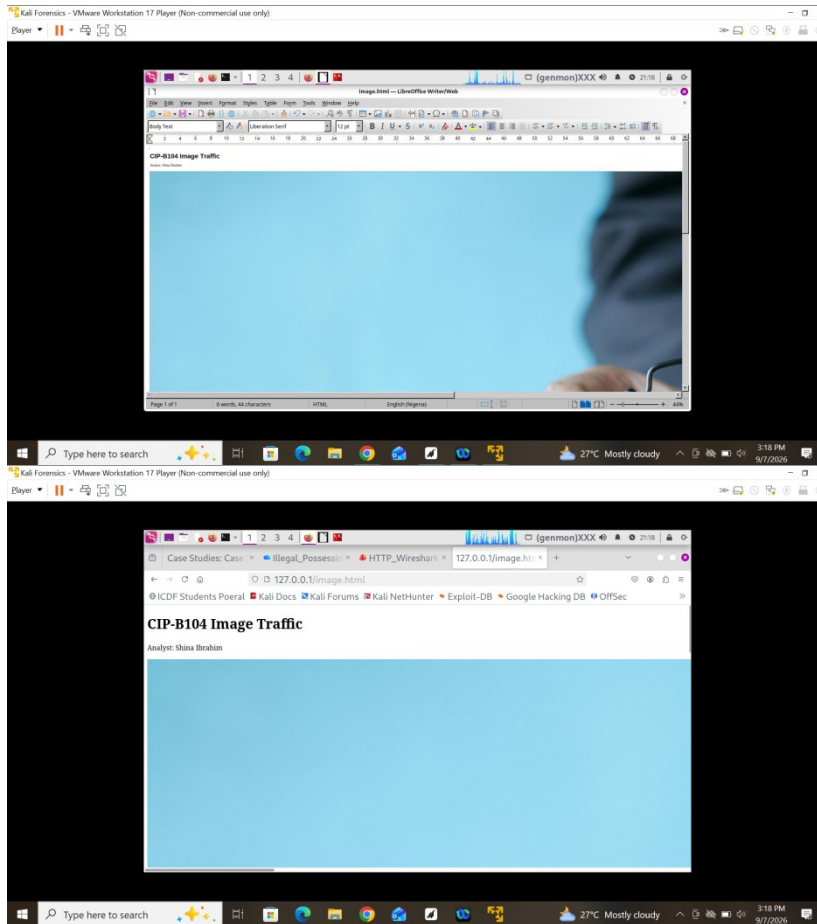




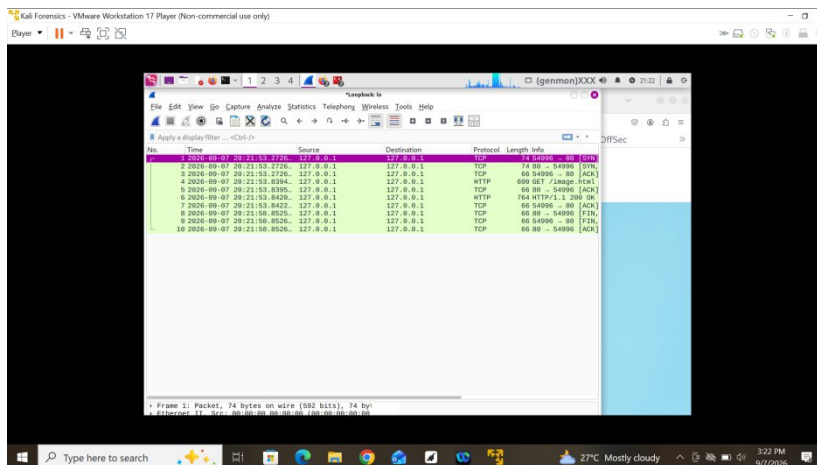
HTTP Wireshark Forensic Image

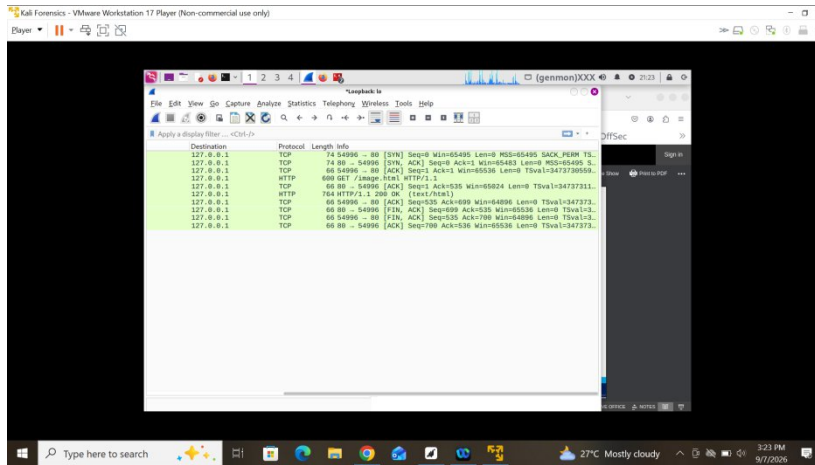
1. Downloaded the image and check its location and also view it on the web



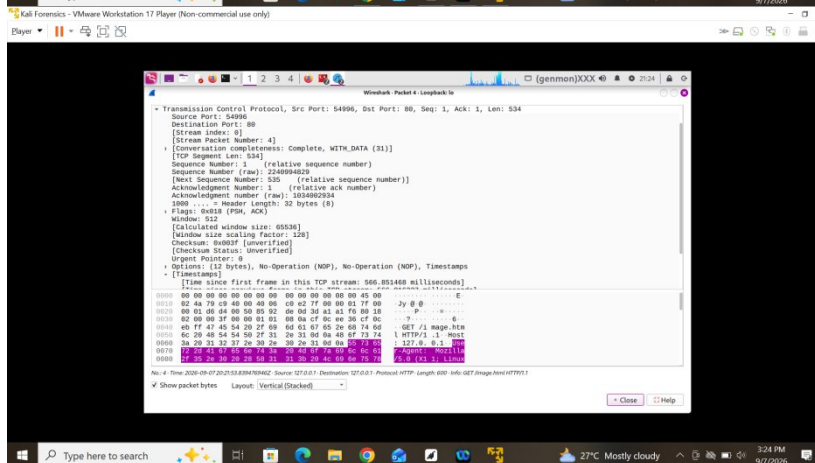
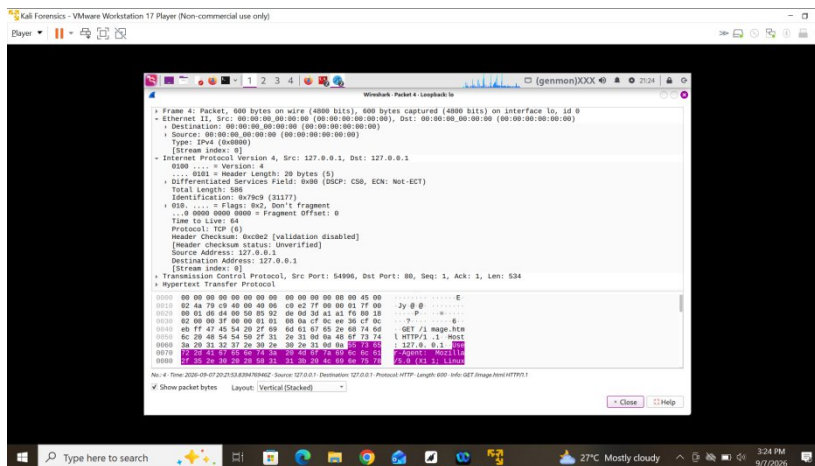


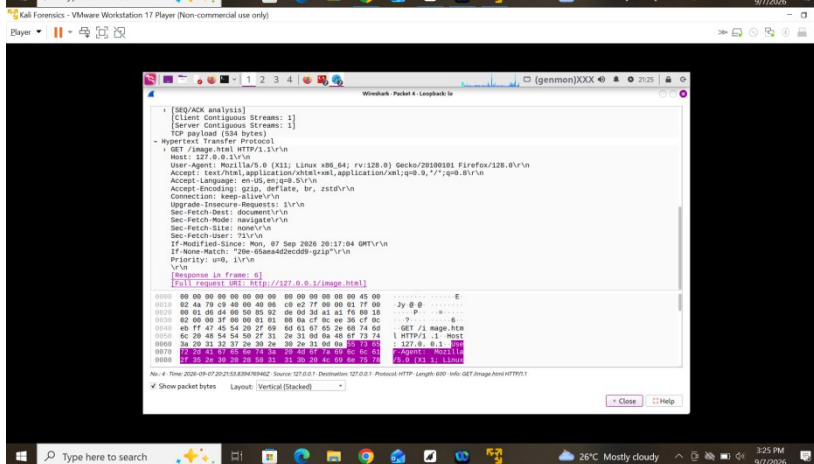
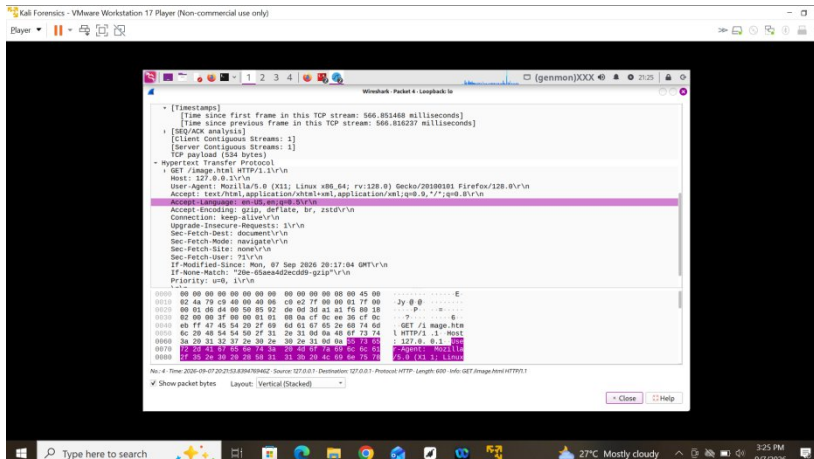
2. This is the image traffic generated



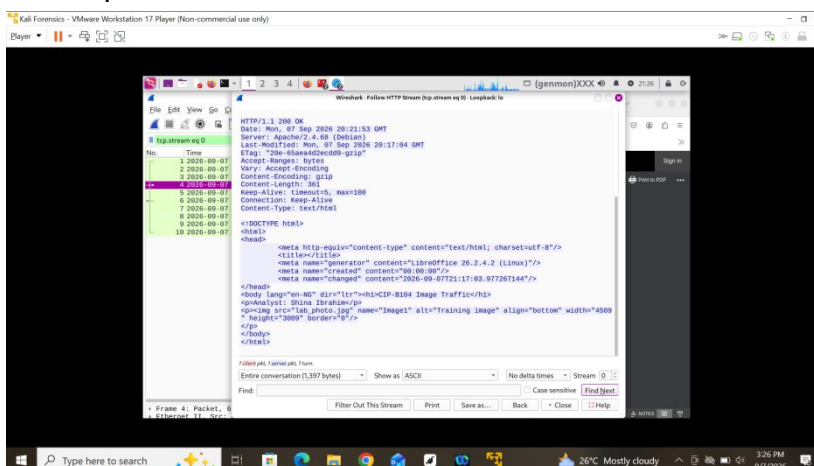


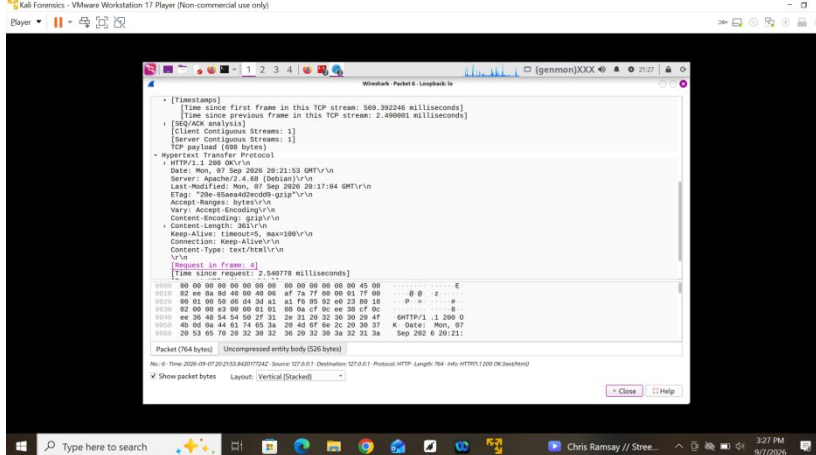
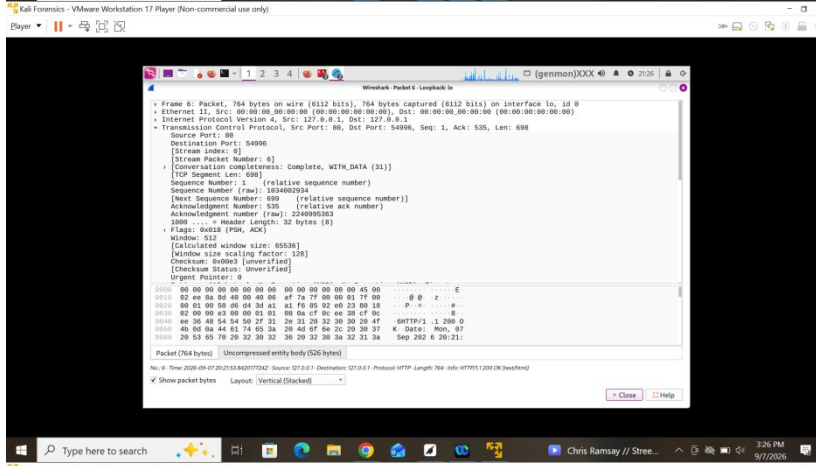
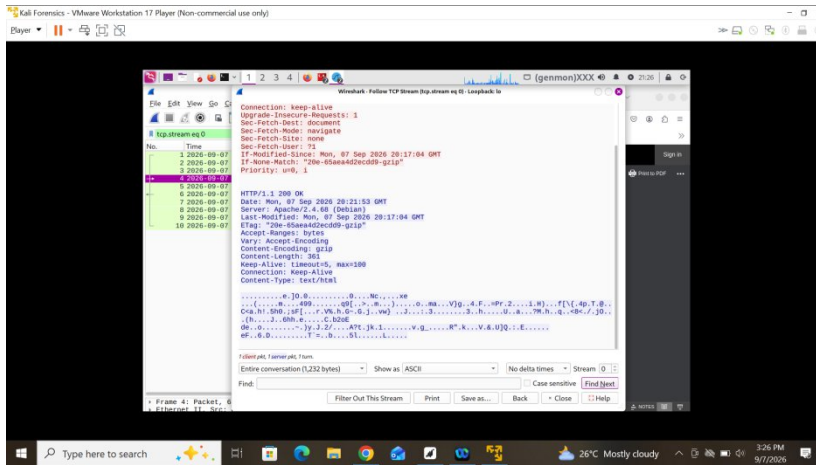
3. Analyzing the traffic by looking t the request, response and following the http stream

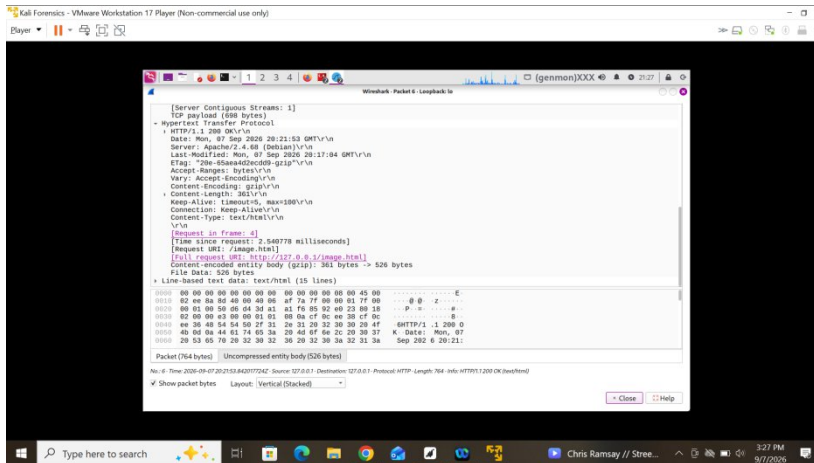




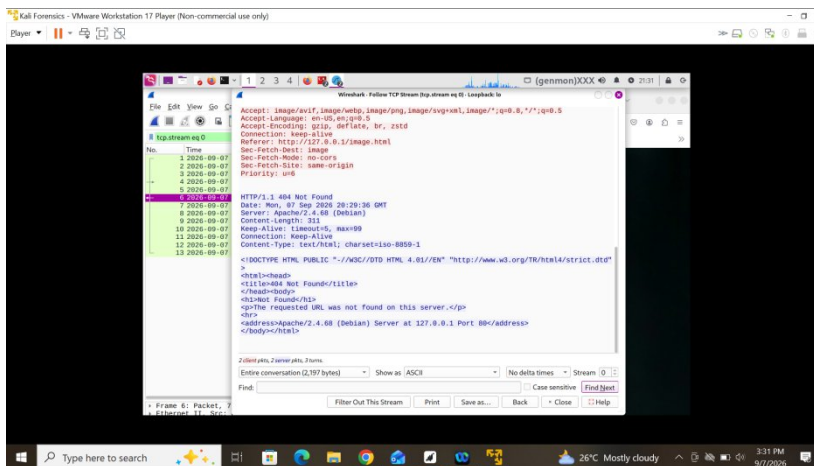
4. The Http Stream

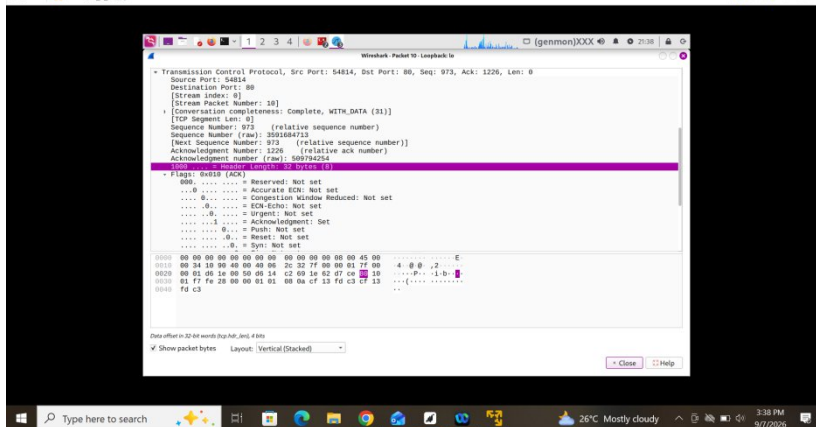
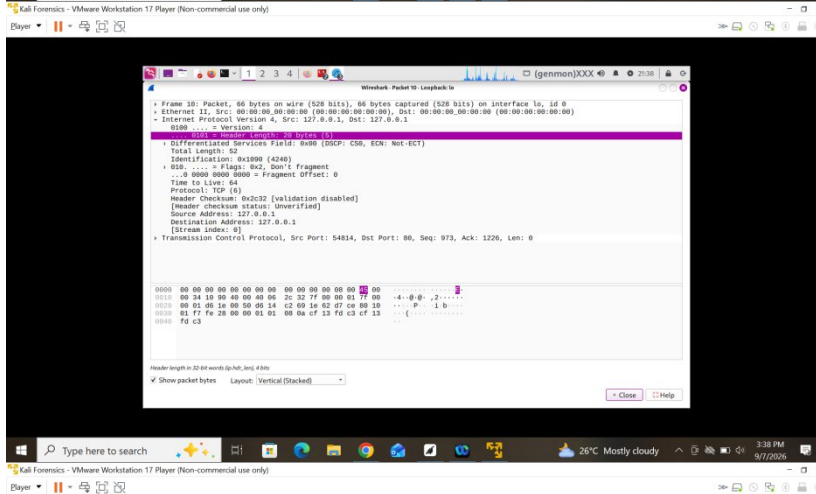
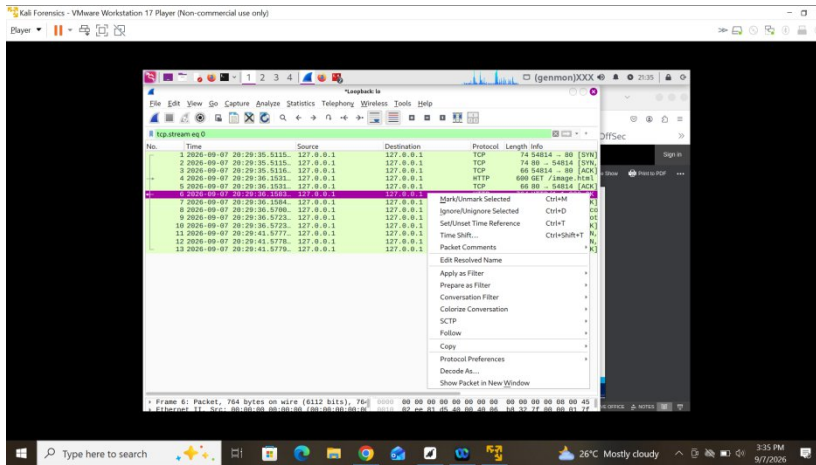




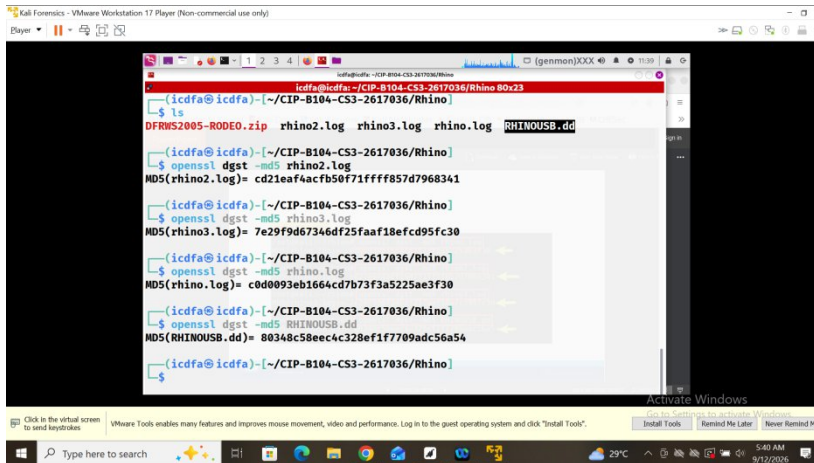


5. The TCP stream

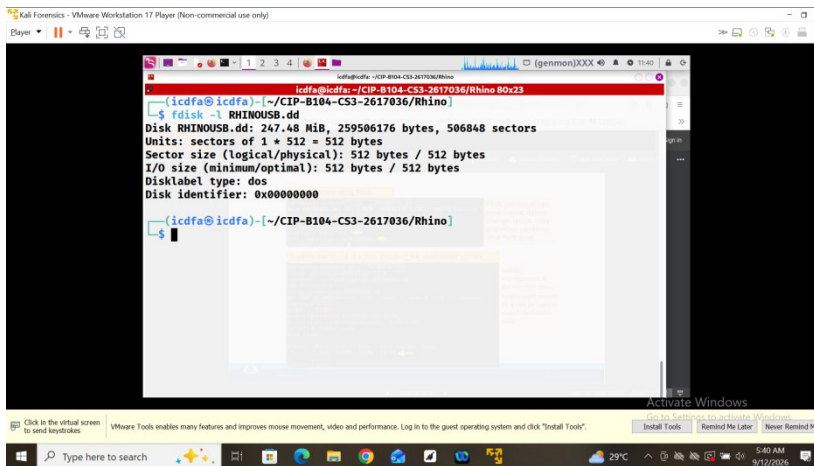




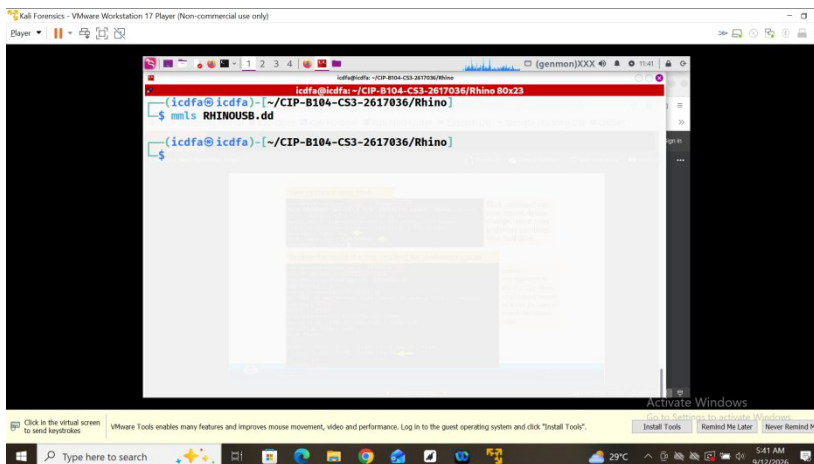
2. Then I looked at the hash value



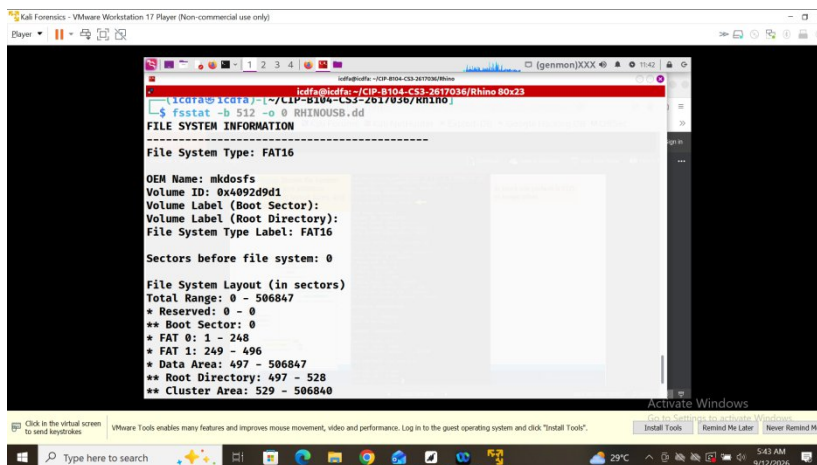
3. Fdisk -l RHINOUSB.dd



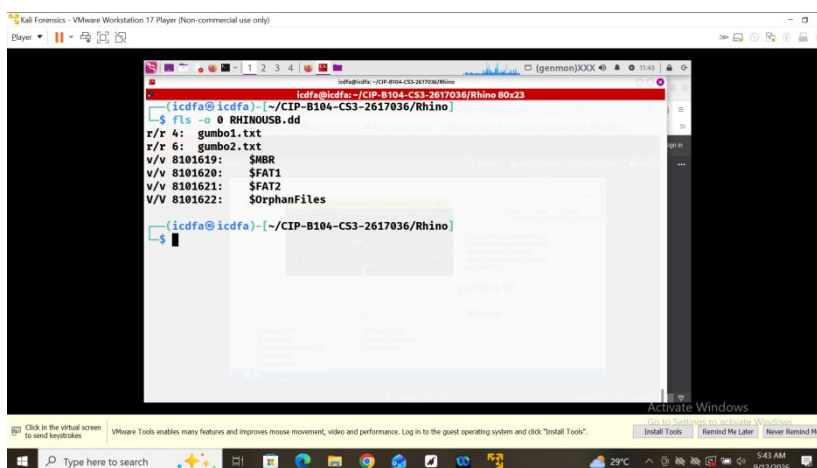
4. Mmls shows no result



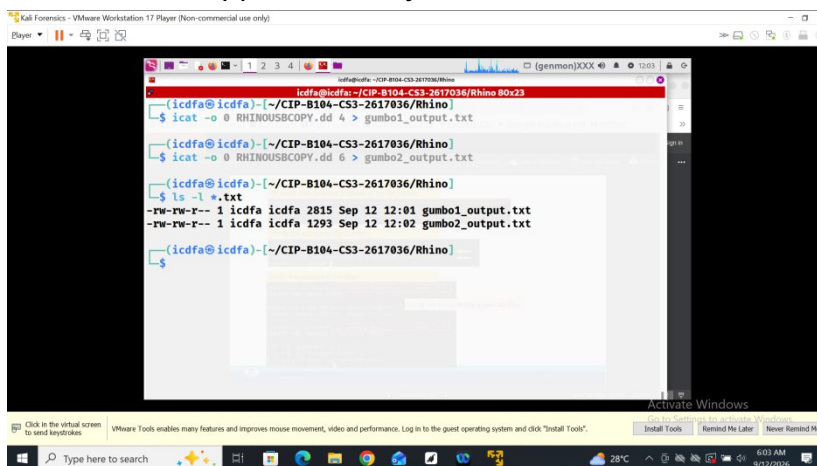
5. Fsstat result



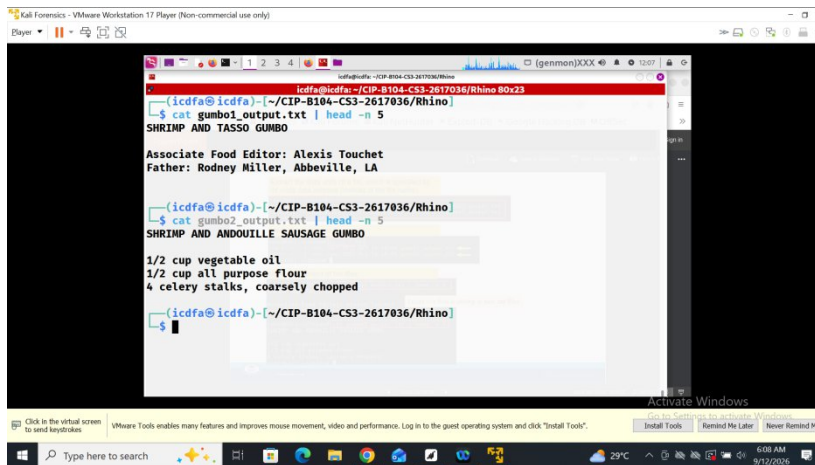
6. Fls result



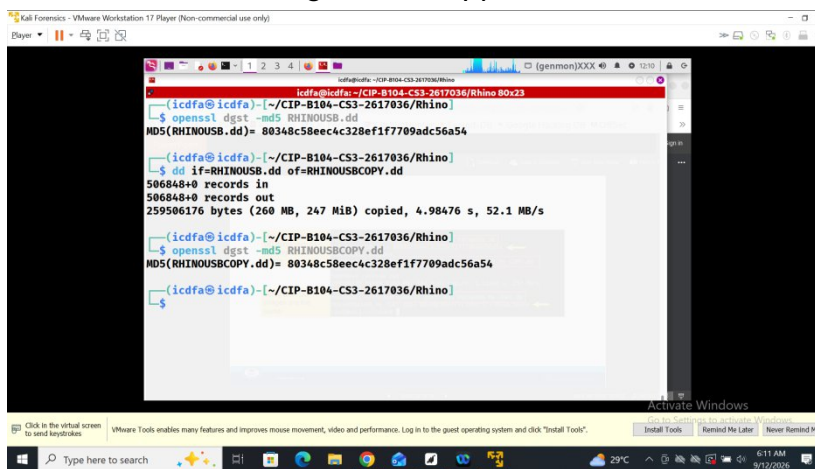
7. Since I have a copy of the file I just ICAT that



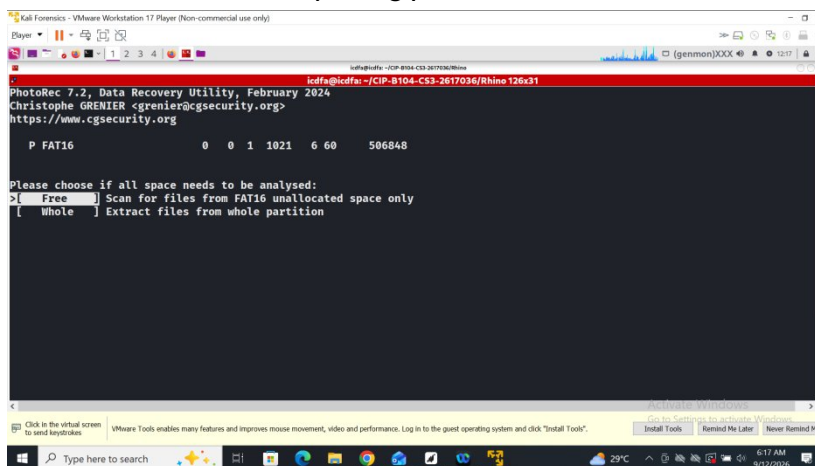
8. I showed the content of both gumbo1 and gumbo2

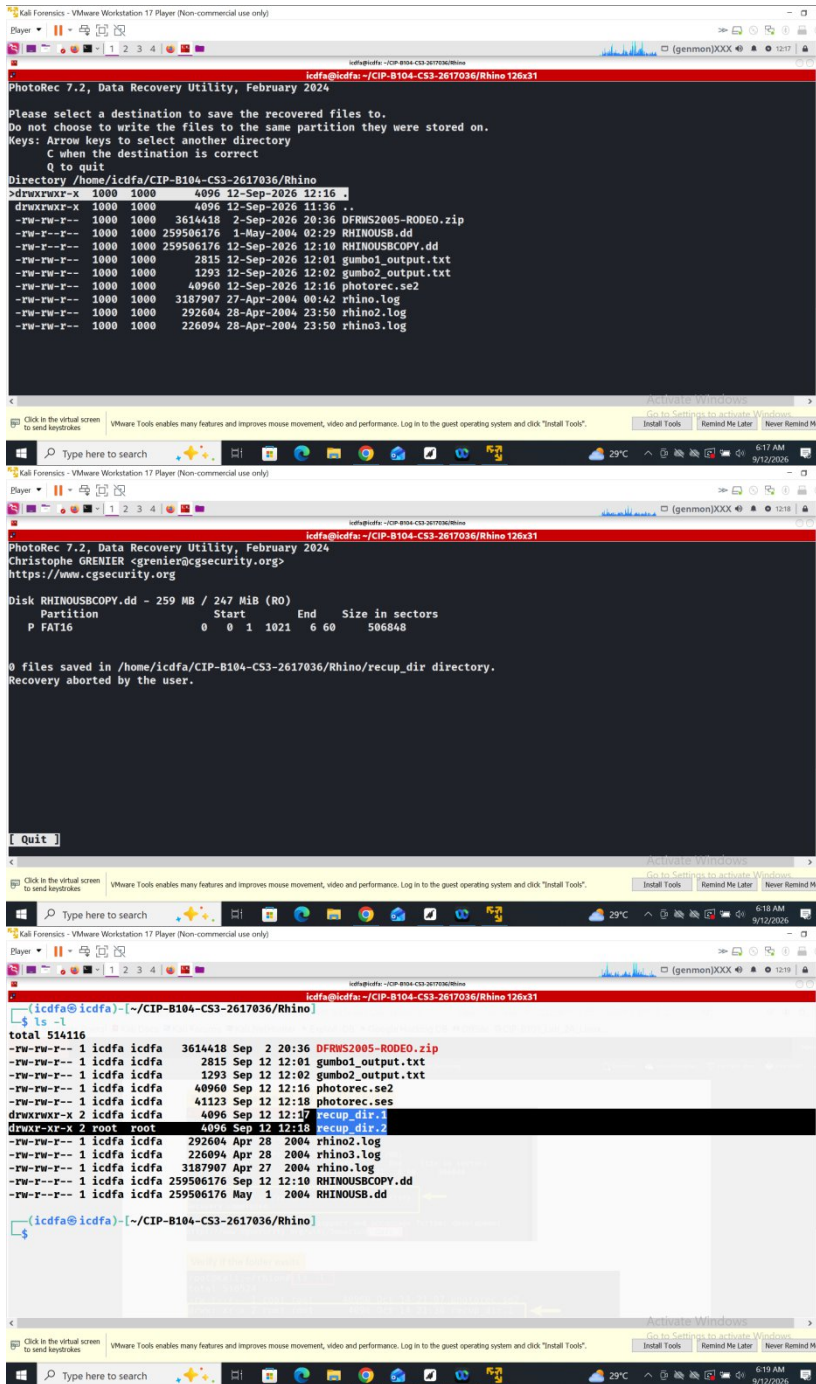


9. I looked at both the original and copy file to make sure the hash value is the same

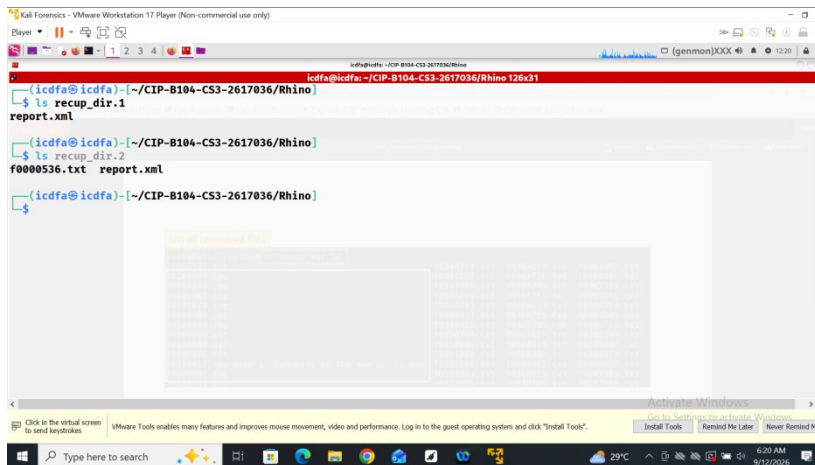


10. Then I start the recovery using photorec





11. The content of both recup1 and recup2



```
icdfa@icdfa:~/CIP-B104-CS3-2617036/Rhino$ ls recup_dir.1
report.xml

icdfa@icdfa:~/CIP-B104-CS3-2617036/Rhino$ ls recup_dir.2
f0000536.txt  report.xml

icdfa@icdfa:~/CIP-B104-CS3-2617036/Rhino$
```

The terminal window shows the output of the 'ls' command for 'recup_dir.1' and 'recup_dir.2'. The output for 'recup_dir.1' is 'report.xml'. The output for 'recup_dir.2' is 'f0000536.txt' and 'report.xml'. Below the terminal output, there is a large block of text, which appears to be the content of the files being analyzed. The text is partially obscured by a watermark and is difficult to read. The terminal window is titled 'Kali Forensics - VMware Workstation 17 Player (Non-commercial use only)'.

Conclusion

The investigation successfully identified, recovered, and correlated rhinoceros images across the USB, network traffic, steganographic files, and archived evidence. Hash comparison was used to eliminate duplicates and maintain accurate identification of unique files. The findings provide reliable evidence of image possession and transfer while maintaining proper forensic integrity and acknowledging the limitations of the available evidence.